

A Trace-Class Bosonic Fock Owner for an Exact Linear Hénon-Type Contraction

Route-A structural certificate C119

Abstract

We freeze a rational linear Hénon-type map whose Euclidean singular values are strictly below one and second-quantize it on standard bosonic Fock space. The resulting operator is trace class. We derive all power traces, an entire Fredholm product, its Taylor coefficients through degree eight, and its full zero divisor. Strict contraction also forces the origin to be the only periodic point. Thus the construction supplies a source-defined analytic operator determinant while failing the nontrivial orbit branch; no external target-divisor or arithmetic correspondence is claimed.

1 Frozen contraction and its orbit obstruction

Let $\Phi(v) = Av$ on $\mathbb{C}^2$, where

$$A = \begin{pmatrix} 3/4 & -1/4 \\ 1/2 & 0 \end{pmatrix}, \quad \chi_A(\lambda) = (\lambda - 1/2)(\lambda - 1/4), \quad \det A = 1/8.$$

This is the recurrence $\Phi(x, y) = (3x/4 - y/4, x/2)$. Euclidean contraction, not merely spectral-radius contraction, follows from

$$A^\top A = \begin{pmatrix} 13/16 & -3/16 \\ -3/16 & 1/16 \end{pmatrix}, \quad s_{1,2}^2 = \frac{7 \pm 3\sqrt{5}}{16}.$$

Both numbers lie in $(0, 1)$: $49 > 45$ gives $7 - 3\sqrt{5} > 0$, while $5 < 9$ gives $7 + 3\sqrt{5} < 16$. Hence $s_1, s_2 < 1$. Moreover, for every $n \geq 1$, the eigenvalues of $A^n - I$ are $2^{-n} - 1$ and $4^{-n} - 1$. Thus $A^n - I$ is invertible, so the origin is the only finite-period point, even over $\mathbb{C}$.

2 A trace-class Fock operator

On the standard bosonic Fock space

$$\mathcal{F}_s(\mathbb{C}^2) = \bigoplus_{m \geq 0} \text{Sym}^m(\mathbb{C}^2), \quad \Gamma(A) = \bigoplus_{m \geq 0} \text{Sym}^m A,$$

the polar decomposition and functoriality of symmetric powers show that the singular values of $\Gamma(A)$ are exactly $s_1^i s_2^j$, $i, j \geq 0$. Consequently

$$\|\Gamma(A)\|_1 = \sum_{i,j \geq 0} s_1^i s_2^j = \frac{1}{(1 - s_1)(1 - s_2)} < \infty.$$

Thus $\Gamma(A)$ is trace class. Since A has distinct eigenvalues, its degree- m symmetric power has eigenvalues $2^{-i}4^{-j}$ for $i + j = m$. Across all degrees, each pair $(i, j) \in \mathbb{Z}_{\geq 0}^2$ occurs once, and therefore, for every $n \geq 1$,

$$t_n := \text{Tr } \Gamma(A)^n = \sum_{i,j \geq 0} 2^{-in} 4^{-jn} = \frac{1}{(1 - 2^{-n})(1 - 4^{-n})}. \quad (1)$$

3 Fredholm product, coefficients, and zeros

Trace class makes the Fredholm determinant entire, and its eigenvalue product is

$$D(z) = \det(I - z\Gamma(A)) = \prod_{i,j \geq 0} (1 - z 2^{-i} 4^{-j}). \quad (2)$$

The product is genus zero because $\sum_{i,j} 2^{-i} 4^{-j} < \infty$. Writing $D(z) = \sum_{n \geq 0} d_n z^n$, the exact recurrence

$$d_0 = 1, \quad n d_n = - \sum_{k=1}^n t_k d_{n-k}$$

gives the independently checked prefix below.

n	t_n	d_n
0	—	1
1	8/3	−8/3
2	64/45	128/45
3	512/441	−32768/19845
4	4096/3825	3014656/5060475
5	32768/31713	−23119003648/160482843675
6	262144/257985	112877646118912/4600240713943875
7	2097152/2080641	−29195669458808471552/9571449439300898023875
8	16777216/16711425	35666710678759844872192/125452987800916870398929625

Equation (2) also gives the complete zero divisor. A zero 2^k arises from $i + 2j = k$; the nonnegative solutions are $(i, j) = (k - 2j, j)$ for $0 \leq j \leq \lfloor k/2 \rfloor$. Hence

$$\text{ord}_{z=2^k} D = \lfloor k/2 \rfloor + 1 \quad (k \geq 0),$$

and there are no other zeros.

4 Reproducibility and Route-A boundary

From the package directory run

```
python3 code/c119_fock_producer.py
python3 code/c119_fock_checker.py
python3 code/c119_sympy_crosscheck.py
python3 code/c119_replay.py
python3 code/c119_mutation.py
```

The independent checker, 21 symbolic checks, byte replay, and all 12 hostile mutations pass. No citation or external data enters the certificate.

The strict evaluator tuple is

`(A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL)`.

A1 records the absence of nontrivial periodic orbits. Although the Fock determinant theorem is exact, A2 fails because this owner is not derived from primitive orbits and no target divisor is matched. A3 fails because no target functional equation, Gamma factor, counting law, or continuation comparison is attempted. Overall status remains Route-A exploratory. We claim no orbit-derived determinant, arithmetic/local data, Euler factors, root numbers, automorphy, Hilbert–Pólya operator, Riemann-zero correspondence, or Route B. The literal firewall is `NO_BAD_EULER_OR_ROOT_NUMBER`.